

Library Management System

You are going to build a project based on Library Management System. It keeps track of all information about books in the library, their cost, status and total number of books available in the library.

Create a database named library and following TABLES in the database:

1. Branch
2. Employee
3. Books
4. Customer
5. IssueStatus
6. ReturnStatus

Attributes for the tables:

1. Branch

- Branch_no
- Set as PRIMARY KEY
- Manager_Id
- Branch_address
- Contact_no

2. Employee

- Emp_Id – Set as PRIMARY KEY
- Emp_name
- Position
- Salary
- Branch_no
- Set as FOREIGN KEY and it refer Branch_no in Branch table

3. Books

- ISBN
- Set as PRIMARY KEY
- Book_title
- Category

- Rental_Price
- Status [Give yes if book available and no if book not available]
- Author
- Publisher

4. Customer

- Customer_Id
- Set as PRIMARY KEY
- Customer_name
- Customer_address
- Reg_date

5. IssueStatus

- Issue_Id
- Set as PRIMARY KEY
- Issued_cust – Set as FOREIGN KEY and it refer customer_id in CUSTOMER table
- Issued_book_name
- Issue_date
- Isbn_book – Set as FOREIGN KEY and it should refer isbn in BOOKS table

6. ReturnStatus

- Return_Id
- Set as PRIMARY KEY
- Return_cust
- Return_book_name
- Return_date
- Isbn_book2
- Set as FOREIGN KEY and it should refer isbn in BOOKS table

- `CREATE DATABASE library;`
- `USE library;`
-  `CREATE TABLE Branch (
 Branch_no INT PRIMARY KEY,
 Manager_Id INT,
 Branch_address VARCHAR(200),
 Contact_no VARCHAR(15)
);`

- `DESC Branch;`

-  `CREATE TABLE Employee (
 Emp_Id INT PRIMARY KEY,
 Emp_name VARCHAR(100),
 Position VARCHAR(50),
 Salary DECIMAL(10,2),
 Branch_no INT,
 FOREIGN KEY (Branch_no)
 REFERENCES Branch(Branch_no)
);`

-  `CREATE TABLE Books (
 ISBN VARCHAR(20) PRIMARY KEY,
 Book_title VARCHAR(200),
 Category VARCHAR(50),
 Rental_Price DECIMAL(10,2),
 Status VARCHAR(5),
 Author VARCHAR(100),
 Publisher VARCHAR(100)
);`

- `DESC Books;`

-  `CREATE TABLE Customer (
 Customer_Id INT PRIMARY KEY,
 Customer_name VARCHAR(100),
 Customer_address VARCHAR(200),
 Reg_date DATE
);`

- `DESC Customer;`

```
CREATE TABLE IssueStatus (  
    Issue_Id INT PRIMARY KEY,  
    Issued_cust INT,  
    Issued_book_name VARCHAR(200),  
    Issue_date DATE,  
    Isbn_book VARCHAR(20),  
  
    FOREIGN KEY (Issued_cust)  
    REFERENCES Customer(Customer_Id),  
  
    FOREIGN KEY (Isbn_book)  
    REFERENCES Books(ISBN)  
);  
DESC IssueStatus;
```

```
CREATE TABLE ReturnStatus (  
    Return_Id INT PRIMARY KEY,  
    Return_cust VARCHAR(100),  
    Return_book_name VARCHAR(200),  
  
    Return_cust VARCHAR(100),  
    Return_book_name VARCHAR(200),  
    Return_date DATE,  
    Isbn_book2 VARCHAR(20),  
  
    FOREIGN KEY (Isbn_book2)  
    REFERENCES Books(ISBN)  
);  
DESC ReturnStatus;
```

70

```
71 • INSERT INTO Branch VALUES
72   (1,101,'Kochi','9876543210'),
73   (2,102,'Thrissur','9876543211'),
74   (3,103,'Calicut','9876543212');
75 • SELECT * FROM Branch;
```

Result Grid				
Filter Rows:				
Edit: 				
	Branch_no	Manager_Id	Branch_address	Contact_no
▶	1	101	Kochi	9876543210
	2	102	Thrissur	9876543211
	3	103	Calicut	9876543212
✱	NULL	NULL	NULL	NULL

```
• INSERT INTO Employee VALUES
  (101,'Rahul','Manager',70000,1),
  (102,'Anjali','Manager',68000,2),
  (103,'Arun','Manager',72000,3),
  (104,'Nikhil','Assistant',45000,1),
  (105,'Meera','Librarian',55000,1),
  (106,'Akhil','Assistant',40000,2),
  (107,'Sneha','Librarian',52000,2),
  (108,'Vivek','Assistant',48000,3),
  (109,'Neethu','Librarian',60000,3);
• SELECT * FROM Employee;
```



```

99 • INSERT INTO Customer VALUES
100 (1,'Asha','Kochi','2021-05-10'),
101 (2,'Ravi','Thrissur','2023-01-15'),
102 (3,'Maya','Calicut','2020-08-20'),
103 (4,'Arjun','Kochi','2022-09-01'),
104 (5,'Diya','Kannur','2021-12-15');
105 • SELECT * FROM Customer;

```

Result Grid				
Filter Rows:				
Edit:   				
	Customer_Id	Customer_name	Customer_address	Reg_date
▶	1	Asha	Kochi	2021-05-10
	2	Ravi	Thrissur	2023-01-15
	3	Maya	Calicut	2020-08-20
	4	Arjun	Kochi	2022-09-01
	5	Diya	Kannur	2021-12-15
*	NULL	NULL	NULL	NULL

```

107 • INSERT INTO IssueStatus VALUES
108 (1,1,'History of India','2023-06-10','ISBN001'),
109 (2,2,'Database Systems','2023-05-20','ISBN002'),
110 (3,3,'World History','2023-06-18','ISBN004'),
111 (4,4,'Python Basics','2024-01-05','ISBN003');
112 • SELECT * FROM IssueStatus;

```

Result Grid					
Filter Rows:					
Edit:    Export					
	Issue_Id	Issued_cust	Issued_book_name	Issue_date	Isbn_book
▶	1	1	History of India	2023-06-10	ISBN001
	2	2	Database Systems	2023-05-20	ISBN002
	3	3	World History	2023-06-18	ISBN004
	4	4	Python Basics	2024-01-05	ISBN003
*	NULL	NULL	NULL	NULL	NULL


```

114 • INSERT INTO ReturnStatus VALUES
115     (1,'Asha','History of India','2023-07-01','ISBN001'),
116     (2,'Ravi','Database Systems','2023-06-01','ISBN002');
117 • SELECT * FROM ReturnStatus;
118

```

Return_Id	Return_cust	Return_book_name	Return_date	Isbn_book2
1	Asha	History of India	2023-07-01	ISBN001
2	Ravi	Database Systems	2023-06-01	ISBN002
NULL	NULL	NULL	NULL	NULL

Display all the tables and Write the queries for the following :

1. Retrieve the book title, category, and rental price of all available books.

```

119 • SELECT Book_title, Category, Rental_Price
120 FROM Books
121 WHERE Status='yes';

```

Book_title	Category	Rental_Price
History of India	History	30.00
Database Systems	Education	40.00
World History	History	28.00
Science Today	Science	20.00
Ancient History	History	32.00

2. List the employee names and their respective salaries in descending order of salary.

```

123 • SELECT Emp_name, Salary
124 FROM Employee
125 ORDER BY Salary DESC;

```

126

```

127 • SELECT b.Book_title,

```

<

Result Grid   Filter Rows:

	Emp_name	Salary
▶	Arun	72000.00
	Rahul	70000.00
	Anjali	68000.00
	Neethu	60000.00
	Meera	55000.00
	Sneha	52000.00
	Vivek	48000.00
	Nikhil	45000.00
	Akhil	40000.00

3. Retrieve the book titles and the corresponding customers who have issued those books.

```

127 • SELECT b.Book_title,
128           c.Customer_name
129 FROM IssueStatus i
130 JOIN Books b
131 ON i.Isbn_book=b.ISBN
132 JOIN Customer c
133 ON i.Issued_cust=c.Customer_Id;

```

<

Result Grid   Filter Rows:

	Book_title	Customer_name
▶	History of India	Asha
	Database Systems	Ravi
	World History	Maya
	Python Basics	Arjun

4. Display the total count of books in each category.

```

134
135 • SELECT Category,
136         COUNT(*) AS Total_Books
137     FROM Books
138     GROUP BY Category;

```

Result Grid		Filter Rows:
Category	Total_Books	
History	3	
Education	2	
Science	2	

5. Retrieve the employee names and their positions for the employees whose salaries are above Rs.50,000.

```

140 • SELECT Emp_name,
141         Position
142     FROM Employee
143     WHERE Salary > 50000;
144

```

Result Grid		Filter Rows:
Emp_name	Position	
Rahul	Manager	
Anjali	Manager	
Arun	Manager	
Meera	Librarian	
Sneha	Librarian	
Neethu	Librarian	

6. List the customer names who registered before 2022-01-01 and have not issued any books yet.

```

145 • SELECT Customer_name
146     FROM Customer
147     WHERE Reg_date < '2022-01-01'
148     AND Customer_Id NOT IN
149     (
150     SELECT Issued_cust
151     FROM IssueStatus
152     );

```

Result Grid | Filter Rows:

Customer_name
Diya

7. Display the branch numbers and the total count of employees in each branch.

```

154 • SELECT Branch_no,
155           COUNT(*) AS Employee_Count
156     FROM Employee
157     GROUP BY Branch_no;

```

Result Grid | Filter Rows: | Exp

Branch_no	Employee_Count
1	3
2	3
3	3

8. Display the names of customers who have issued books in the month of June 2023.

```

159 • SELECT DISTINCT c.Customer_name
160 FROM Customer c
161 JOIN IssueStatus i
162 ON c.Customer_Id=i.Issued_cust
163 WHERE MONTH(i.Issue_date)=6
164 AND YEAR(i.Issue_date)=2023;

```

Result Grid		Filter Rows:
	Customer_name	
▶	Asha	
	Maya	

9. Retrieve book_title from book table containing history.

```

166 • SELECT Book_title
167 FROM Books
168 WHERE Book_title LIKE '%History%';

```

Result Grid		Filter Rows:	Exp
	Book_title		
▶	History of India		
	World History		
	Ancient History		

10. Retrieve the branch numbers along with the count of employees for branches having more than 5 employees

```
169
170 • SELECT Branch_no,
171         COUNT(*) AS Employee_Count
172     FROM Employee
173     GROUP BY Branch_no
174     HAVING COUNT(*) > 5;
```

Result Grid | Filter Rows: | Ex

Branch_no	Employee_Count
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11. Retrieve the names of employees who manage branches and their respective branch addresses.

```
176 • SELECT e.Emp_name,
177         b.Branch_address
178     FROM Employee e
179     JOIN Branch b
180     ON e.Emp_Id=b.Manager_Id;
```

< Filter Rows:

	Emp_name	Branch_address
▶	Rahul	Kochi
	Anjali	Thrissur

12. Display the names of customers who have issued books with a rental price higher than Rs. 25.

```
182 • SELECT DISTINCT c.Customer_name
183 FROM Customer c
184 JOIN IssueStatus i
185 ON c.Customer_Id=i.Issued_cust
186 JOIN Books b
187 ON i.Isbn_book=b.ISBN
188 WHERE b.Rental_Price > 25;
```

Result Grid	
Filter Rows:	
Customer_name	
▶ Asha	
Ravi	
Arjun	
Maya	